

Answers to the Exercises

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Answers to the Exercises — Translation

1. $\neg E$
2. $\neg E \rightarrow \neg U$
3. $U \rightarrow F$
4. $\neg U \wedge \neg M$
5. $D \rightarrow M$
6. $\neg S \rightarrow B$
7. $(W \rightarrow T) \rightarrow \neg S$
8. $\neg L \rightarrow (\neg S \rightarrow P)$
9. $(A \rightarrow B) \rightarrow (W \rightarrow B)$
10. $\neg A \wedge \neg B \rightarrow M \rightarrow N$
11. $\neg E \rightarrow (P \wedge \neg K \rightarrow A)$
12. $(\neg R \rightarrow B) \wedge (R \rightarrow P \vee S)$
13. $(S \wedge \neg C \rightarrow B) \wedge (\neg(S \wedge \neg C) \rightarrow P \vee S)$
14. $(E \rightarrow H) \wedge \neg(H \rightarrow E)$

Answers to the Exercises — Translation

1. $\forall x \forall y (Hxy \rightarrow Uxy)$
2. $\forall x (\neg Sx \rightarrow \exists y (Eyx \wedge \neg Sy))$
3. $\forall x (Mx \vee Wx \rightarrow Ax)$
4. $\neg \forall x (Jx \rightarrow Dx)$
5. $\forall x (Jx \rightarrow \neg Dx)$
6. $\forall x (\neg (Rx \wedge Bx) \rightarrow \neg Ex)$
7. $Hs \wedge Hp \wedge \forall x (Hx \rightarrow x = s \vee x = p)$
8. $\forall x (x \neq s \wedge x \neq p \rightarrow Hx)$
9. $\forall x (Bx \rightarrow \exists y \exists z (Gy \wedge Gz \wedge y \neq z \wedge Lxy \wedge Lxz))$
10. $\neg \forall x (\text{Glitter}(x) \rightarrow \text{Gold}(x))$
11. $\forall x y z (\text{City}(z) \wedge \text{In}(x, z) \wedge \text{In}(y, z) \rightarrow \text{first}(\text{code}(x)) = \text{first}(\text{code}(y)))$
12. $\forall x \forall y (Bx \wedge Gy \wedge Fyx \rightarrow Hx)$
13. $\forall x \forall y (Bx \wedge Gy \wedge Fyx \rightarrow Cxy)$
14. $\forall x (Ex \rightarrow Dx2) \wedge \exists x (Ex \wedge Dx4) \wedge \exists x (Ex \wedge \neg Dx4)$

Answers to the Exercises — Translation

15. $\forall x(Bx \wedge \forall y(Gy \rightarrow Lxy) \rightarrow \neg \exists z(Gz \wedge Lzx))$
16. $\forall x(Cx \rightarrow \exists y(Iyx \wedge \forall z(Izx \rightarrow Lzy)))$
17. $\forall x(Dx \rightarrow Ax) \rightarrow \forall x(\exists y(Dy \wedge Hxy) \rightarrow \exists y(Ay \wedge Hxy))$
 $\forall x(Dx \rightarrow Ax) \rightarrow \forall x(Dx \rightarrow \exists y(Ay \wedge hx = hy))$
18. $\forall x(\neg D(x, f(w(s))) \rightarrow x = m(w(s)))$
19. $\forall x \exists y \exists z(Gy \wedge Gz \wedge y \neq z \wedge Lxy \wedge Lxz \rightarrow x = d)$
20. $\forall x(\neg Lxx \rightarrow Lhx)$
21. $\exists x(Gx \wedge Lxq \wedge \forall y(Gy \wedge Lyq \rightarrow y = x) \wedge Lqx \wedge \forall y(Lqy \rightarrow y = x))$
22. $\exists x(Lxw \wedge \forall y(Lyw \rightarrow y = x) \wedge \exists y(Lwy \wedge \forall z(Lwz \rightarrow z = y) \wedge x \neq y))$
23. $\exists x(Bx \wedge \forall z(Bz \rightarrow Txz) \wedge \exists y(Gy \wedge \forall z(Gz \rightarrow Tzy) \wedge Lxy))$
24. $\forall x(Cx \wedge Ex \rightarrow \neg \exists y(Py \wedge Sy) \rightarrow Dx)$
25. $\forall x(Bx \rightarrow \neg \exists y(Ty \wedge Axy) \vee \forall y(Ty \rightarrow Axy))$
26. $\forall x(Mx \wedge \exists y \exists z(Cyz \wedge Lxy) \rightarrow \exists y Cyx)$
27. $\exists x \exists y \exists z(Mxa \wedge Myb \wedge Mzx \wedge Mzy) \wedge \forall u \forall v \forall x \forall y(Mua \wedge Mvb \wedge Mxu \wedge Myv \rightarrow x = y)$
 $m(m(a)) = m(m(b))$
28. $\exists x(Gx \wedge Lxb \wedge \forall y(Gy \wedge Lyb \rightarrow y = x) \wedge \exists y(y \neq x \wedge Sy))$

Answers to the Exercises — Validity

1. $\exists x Jx, \neg \exists x (Ax \wedge Jx) \vdash \exists x \neg Ax$
2. $\forall x (Cx \vee Ax \rightarrow Lxa), \forall x (Zx \rightarrow Cx \vee Ax) \vdash \forall x (Zx \rightarrow Lxa)$
3. $\forall x (\neg Ax \wedge Mx \rightarrow Vx \vee Fx), \forall x (Px \rightarrow \neg Ax \wedge \neg Vx \wedge \neg Fx) \vdash \forall x (Px \rightarrow \neg Mx)$
4. $\forall x (Dx \rightarrow Ax) \rightarrow \forall x (\exists y (Dy \wedge Hxy) \rightarrow \exists y (Ay \wedge Hxy))$
5. $\forall x (Px \rightarrow x = a), Pb \wedge Sb \vdash Sa$
6. $Lab, Lbc, Ma, \neg Mc \vdash \exists x \exists y (Mx \wedge \neg My \wedge Lxy)$
7. $\forall x Lxa, \forall x (Lax \rightarrow x = i) \vdash \forall x Lxi$
8. $\forall x \forall y (\exists z Lyz \rightarrow Lxy) \vdash Lrj \rightarrow Lum$
9. $\neg \exists x (Bx \wedge \forall y (Sxy \leftrightarrow \neg Syy))$
10. $\forall x \forall y (Fx \wedge Fy \wedge Bxy \rightarrow Sxy) \rightarrow \exists x (Fx \wedge \forall y (Fy \rightarrow Bxy)) \rightarrow \exists x (Fx \wedge \forall y (Fy \rightarrow Sxy))$
11. $\forall x (Sx \rightarrow Kx), \exists x Sx, \forall x (Kx \rightarrow x = a) \vdash Sa$

Answers to the Exercises — Validity

- 12. $\neg\exists x\exists y(Gx \wedge Sy \wedge Lxy), Ga \wedge \forall x(Lxa \rightarrow Lax), Lba \vdash \neg Sb$
- 13. $\forall xCx \rightarrow \neg\exists xNx, \forall x\neg Cx \rightarrow \neg\exists xEx \vdash \exists x(Nx \wedge Ex) \rightarrow \exists xCx \wedge \exists x\neg Cx$
- 14. $\neg\exists x\forall y(Txy \leftrightarrow \neg\exists z(Tyz \wedge Tzy))$
- 15. $Ri \wedge Pi, \forall x\forall y(Px \wedge Axy \rightarrow Py), \neg\exists x(Px \wedge \neg Ex), \forall x(Ex \wedge Rx \rightarrow \exists y(Gy \wedge Axy)), \forall x(Ex \rightarrow Sx) \vdash \exists x(Gx \wedge Sx)$
- 16. $\exists x\forall y((Ky \leftrightarrow y = x) \wedge Bx), \forall x(Bx \rightarrow Sx) \vdash \forall x(Kx \rightarrow Sx)$
- 17. $Aac \wedge Abc \wedge a \neq b \wedge \forall x(Axc \rightarrow x = a \vee x = b), \forall x(Axc \leftrightarrow Lxc) \vdash \exists x\exists y(Lxc \wedge Lyc \wedge x \neq y \wedge \forall z(Lzc \rightarrow z = x \vee z = y))$
- 18. $\forall x\forall y(Sxy \rightarrow \exists zIzxy), \forall x\forall y\forall z(Izxy \rightarrow Kzx \wedge Kzy), \forall xSxa \vdash \forall x\exists y(Iyxa \wedge Kya)$